

Computer Science and Engineering
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Artificial Intelligence 2

Practical Work IV

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Abstract

In this practical work, we explored how genetic algorithms can be used to optimize an investment portfolio by maximizing the Sharpe Ratio. Our implementation used genetic operators such as selection, crossover, mutation, and elitism to improve the weights assigned to the different assets of the portfolio over successive generations. The objective was to assess how different configurations, particularly the elitism percentage impacted the algorithm's ability to find efficient portfolios with high return-to-risk ratios. Throughout the process, we evaluated different key metrics such as fitness, best individual, and consensus individuals. These experiments helped us view the behavior of the algorithm and how some small changes in the parameters could affect the convergence.

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1. Introduction

Genetic algorithms are a way to solve problems by simulating the process of natural evolution. In this practical work, we developed a genetic algorithm with the goal of reaching an optimal investment portfolio by adjusting the proportions of capital invested to a set of financial assets. Each individual in the population represents a potential portfolio, where the genes correspond to the weights assigned to each asset. Across multiple generations, the algorithm seeks to improve the performance of the population using selection, crossover, mutation, and elitism.

Our main objective is to understand how different genetic operators and configurations, particularly elitism affect the algorithm's ability to maximize the Sharpe Ratio, a key metric that balances return and risk. We conducted experiments with different elitism percentages and tracked how the population evolved over time in terms of fitness, diversity, and convergence.

The purpose was not only to find the most efficient portfolio configuration, but also to analyze how parameter tuning influences the speed and quality of convergence. This research document reflects what we did and what we learned.

All experiments done are influenced by the learnings obtained in class, with the slides provided by our teacher from Unit 8 – Genetic Algorithms [1] and research of normalization methods, like Unit 8 – Genetic Algorithms

2. Exercise

2.1. Description of the normalization method.

In this exercise, we have implemented a normalization method during several parts of the genetic algorithm, such as initialization, mutation, and crossover. We have chosen to do it with the **Simplex projection normalization method** due to its relevant use in machine learning problems, particularly when dealing with data representing proportions, which is our case.

The way this normalization method works is by normalizing data by projecting it onto a simplex.

A simplex in portfolio optimization refers to all the possible portfolios available having to constraints:

- 1) The sum of all the weights (proportion invested in each asset) is equal to 1.
- 2) The value of the weights are positive.

$$\mathcal{S} = \left\{ w \in \mathbb{R}^n \mid \sum_{i=1}^n w_i = 1, w_i \geq 0 \forall i \right\}$$

Equation 1 – Simplex representation

The process of the normalization method simplex projection is the following:

- 1) Sort the weights vector in descending order.
- 2) Calculate the threshold using the sorted weights of the portfolio.
- 3) Subtract the threshold to the original weights and take the maximum of that value and 0, obtaining the normalized weights.

This normalization technique ensures that after the initialization and executing the mutation and crossover operators during the genetic algorithm process, the different weights sum 1 and all of them are positive,

2.2. Description of the experimentation process conducted for optimization

In this practical work the dataset was composed of 20 different assets of different companies and their monthly returns for 1 year. The data preparation was very simple, the only modification done was to transpose the dataset, so the names of the assets were in the first row, and the months in the first column.

After the data preparation we build all the genetic algorithm process, such as defining the chromosome, the population, the normalization method, the different operators (selection, mutation and crossover), the fitness function (Sharpe ratio), the different hyperparameters... for a good performance of the genetic algorithm and optimize the investment portfolio.

During the experimentation phase, we ran the genetic algorithm using different configurations to see how elitism affects the algorithm and find the best portfolio possible. The elitism percentages values uses have been: 0.05, 0.1, 0.2, 0.3 and 0.4 with the same parameters for all of them to see the performance of all of them and evaluate which percentage was the best.

Throughout the experiment, we tested different values for mutation and crossover probabilities, and generations, always comparing the results carefully. Some of the configurations we have tried are the following:

1st configuration:

- Population size: 1000
- Generations: 10000
- Crossover probability (Pc): 0.1
- Mutation probability (Pm): 0.8
- nres: 20 generations
- nsam: every 150 generations

2nd configuration:

- Population size: 1000
- Generations: 10000
- Crossover probability (Pc): 0.05
- Mutation probability (Pm): 0.9
- nres: 2 generations
- nsam: every 150 generations

After some generations, we recorded the measurements specified in the practical work instructions:

BESTf: The best fitness value in the population

BESTn: How many individuals had the best fitness

BEST%: Percentage of individuals with the best fitness

Every 150 generations, we also took a random sample of 20% of the population to check the diversity of the portfolios, how were distributed the weights and an overview of their performance.

After executing the different elitism percentages with different configurations, we evaluated the best elitism percentage, and we made multiple graphs showing how the following metrics changed over time. These graphs helped us to identify patterns and see the effect of elitism on how quickly and well the population converged.

The metrics analyzed for the best elitism percentage are: BESTf, BESTn, and BEST%. They have been analyzed with respect to the epochs.

2.3. Questions

2.3.1. What type of normalization have you used to ensure that the sum of all genes is 1 and that there are no negative values?

The type of normalization used for solving this problem and ensuring that the weights of the different assets of the portfolio are positive, and the sum of all the weights is equal to 1, is the **simplex projection normalization**, explained before, in the section 2.2. After initialization and each genetic operation (crossover, and mutation), every chromosome (portfolio) is normalized by projecting its weight vector onto the standard simplex.

This guarantees that every chromosome represents a valid portfolio due to:

- All genes (weights) are non-negative.
- The sum of all genes is exactly 1.

Throughout the experiment, we tested different values for population size, mutation probabilities, and generations, always comparing the results carefully. The combination we presented in this report, with elitism and the selected parameters, was the best and most stable we found after several tests.

2.3.2. Try different elitism percentages and determine which one performs best percentage. Create a comparison table and explain the details of the chosen configuration.

Metric	Elitism Percentage (5%)	Elitism Percentage (10%)	Elitism Percentage (20%)	Elitism Percentage (30%)	Elitism Percentage (40%)
BESTf	0.4510	0.4680	0.4634	0.4665	0.4659
BESTn	72	131	241	350	453
BEST%	7.20%	13.10%	24.10%	35%	45.30%

Table 1 – Elitism comparison

After trying different elitism percentages (5%, 10%, 20%, 30% y 40%) we have evaluated their performance through different metrics, like the best fitness achieved (BESTf), the number of portfolios with the best fitness value (BESTn), and the percentage of individuals it represents (BEST%). Through the different metrics we have observed that **the one with 10% is the elitism percentage with the best performance**, achieving the best fitness value of 0.4680, which is a good value for a Sharpe ratio (taking into account that the data were monthly returns). Also, the value of number of instances with the best fitness is 131, which a good value, representing 13,10% of the population has the highest Sharpe ratio. These values indicate the algorithm has achieved a balance between exploiting and exploring, maintaining good individuals at the same time as ensuring diversity. Also, this allows to not achieve an early convergence and getting stuck in local optima.

The configuration established for achieving the best elitism percentage was the following:

n=1000, generations=1000, Pc=0.8, Pm=0.1, nres=20, nsam=120

This configuration established is due to several reasons. A population of 1000 individuals is big enough to maintain diversity and reduce early convergence. 1000 generations allows the algorithm to have time to be optimal. A crossover probability of 0.8 allows to explore new portfolio individuals and don't get stuck. And a mutation probability of 0.1 helps to maintain diversity and don't achieve a local optima too fast.

We can see in the following image the evolution of the portfolio performance measured with the Sharpe ratio across the different generations. We can see how the elitism with a value of 10% improves constantly, and doesn't converges too fast.

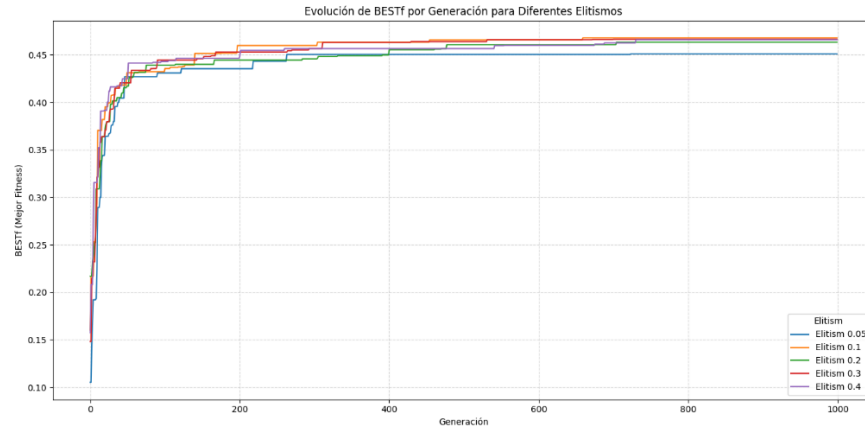


Figure 1 - BESTf vs Generation (Different elitism percentages)

2.3.3. For the best percentage analyze the evolution of the metrics (BESTf, BESTn, BEST%) with respect to the number of epochs, and n for different executions. Use different graphs to illustrate the relationships between these variables, and attempt to identify patterns or correlations that lead to the best results. Provide a detailed explanation of your findings and what occurs during the evolution process.

For the best percentage analyzed, the one with a value of 10%, we have done different graphs to visualize the results achieved in different metrics.

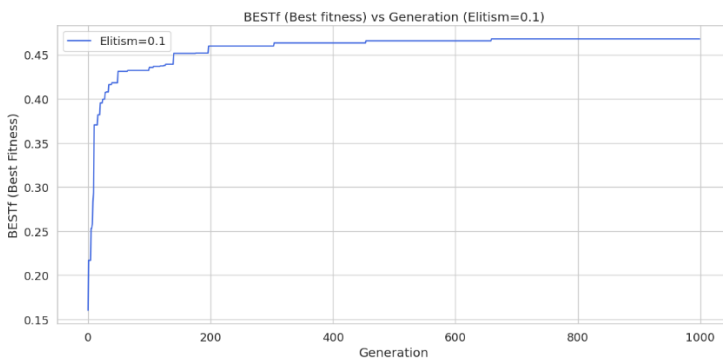


Figure 2 - BESTf vs Generation (Elitism = 0.1)

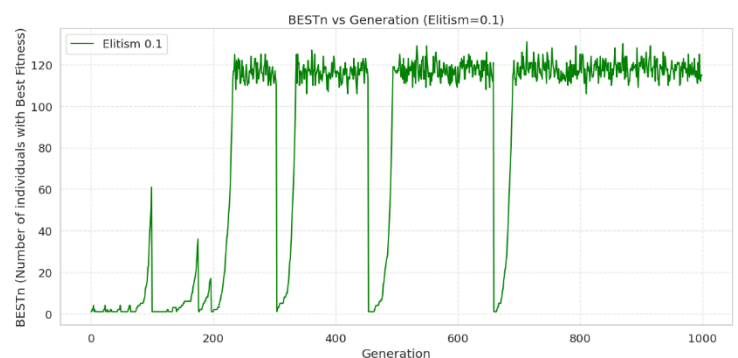


Figure 3 - BESTn vs Generation (Elitism = 0.1)

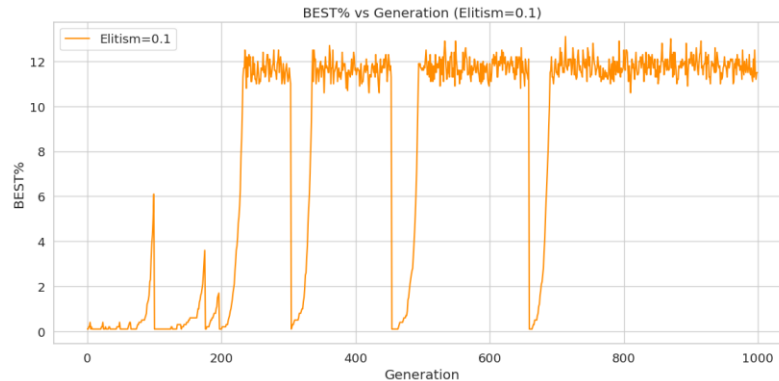


Figure 4 - BEST% vs Generation (Elitism = 0.1)

After representing different metrics for the elitism percentage of 10% we can see how the algorithm works through the different generations. For the first metric, the BESTf we can see how the population learns very fast obtaining a high level of fitness, and then the improvements are minimal, although it never stops in a local minimum. The process of learning through exploration and exploitation has been successfully achieved.

For the second and third metric we can see the representation is very similar. Initially the BESTn and BEST% is low and fluctuates significantly. Then we can see it maintains at a high level, in general is stable, although in some points there are drops that recover fast.

2.3.4. For the best percentage, what is the maximum value that the fitness can reach? Why?

For the best percentage, 10%, the maximum value reached is 0.4680. This value is good, although it could be better. Based on the different configurations we have tried is the best fitness value we have obtained. Considering that our risk-free is 0.003757, a positive value means that the portfolio is generating good returns relative to the risk it is taking. The higher the value, the better the return-to-risk ratio. This means that our optimal portfolio is expected to generate an excess return of 0.4680 units for every unit of risk taken.

This Sharpe value is due to different reasons. Firstly, because of the data provided, the period time (one year) influences in terms of an optimal portfolio, as well as the monthly returns of the different assets. Also, the parameters used have an impact on the results obtained. Trying more different values for the parameters with the different elitism percentages: population size, number of generations, crossover probability and mutation probability, could have helped achieve a better fitness value.

3. Conclusions

To conclude all the work we've done, we learned a lot about how genetic algorithms behave and how important the right configuration is.

We have been able to try and evaluate different elitism percentages, which is a very important parameter to help the algorithm to perform better. Also, another realized small changes, like adjusting different parameters, can make a big difference.

In our case, with the parameters applied and the data used, we have achieved that the best elitism for our project is 10%, we have get good results obtaining a best fitness value of 0.4680. This is the result of a process of testing different parameters to get an algorithm that converges in an optimal way, learning generation by generating through exploring and exploiting.

Overall, this project showed us how important is to test, observe, and improve step by step, to get good results.

4. References

- [1] M. M. Muñoz, “Unit 8. Genetic Algorithms”.